

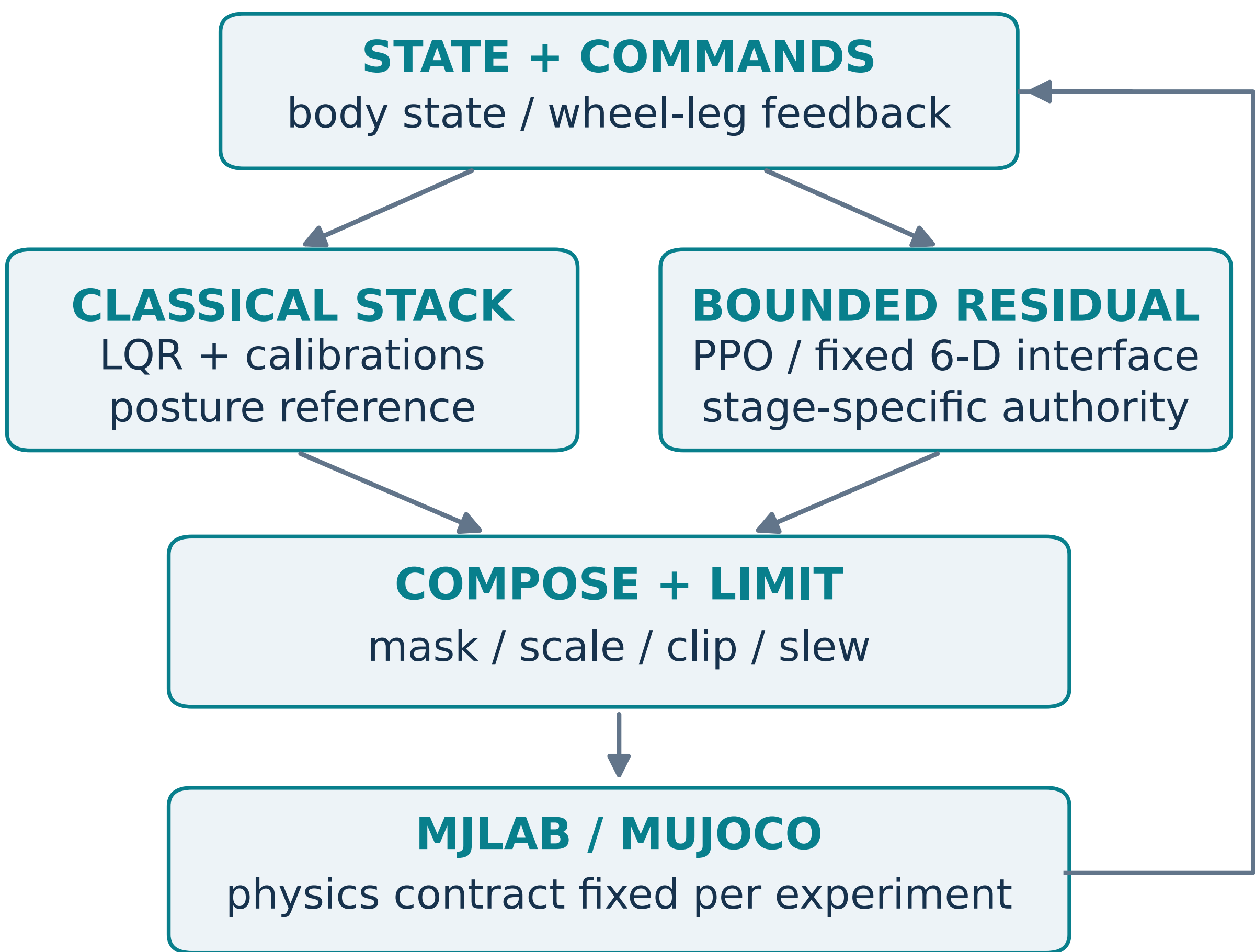
Engineering a Traceable Hybrid Controller for a Two-Wheeled Legged Robot

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01 System architecture

Balance, motion and disturbance recovery share the same wheels. Four leg joints change the posture and the balance operating point. The engineering task is to make these responsibilities explicit and testable.



Zero policy output = the complete classical baseline

Fixed interface, controlled authority. The policy always has six outputs: balance, yaw and four leg corrections. Stage masks select the permitted channels; limits constrain the applied targets.

The recovery experiment used its earlier frozen classical stack. The later C1 experiment qualified an affine gain schedule. These are separate versions, not one interchangeable baseline.

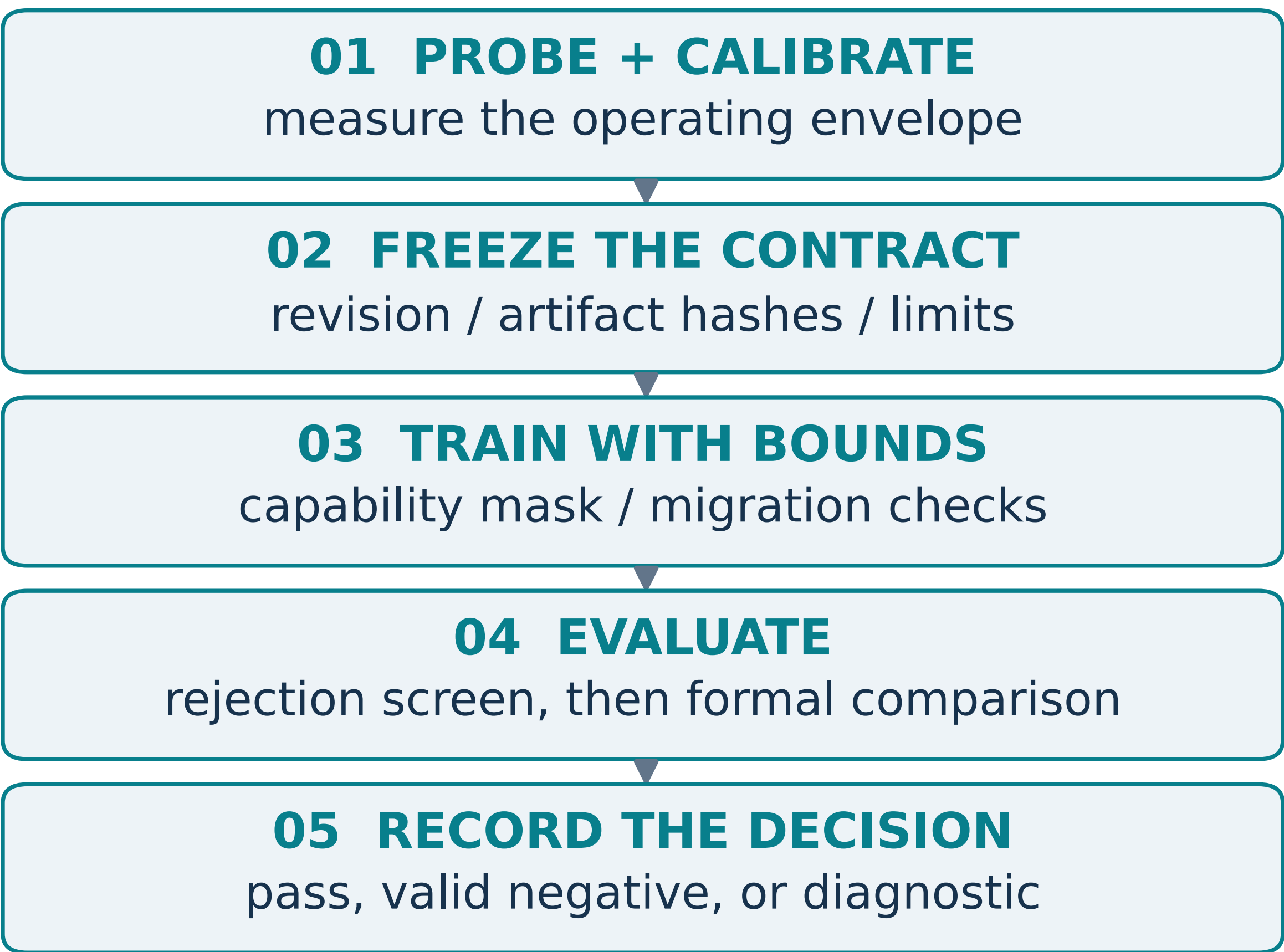
What was engineered

- **Control:** identified LQR, calibrated velocity/yaw response, posture mapping and rate-limited references.
- **Learning:** bounded residual PPO, migration checks and checkpoint-to-artifact binding.
- **Interfaces:** portable classical control, hardware abstractions, mock motor buses and a latching safety supervisor.

Using PPO or MuJoCo is not claimed as a new algorithm. The contribution is their integration into an inspectable control and evaluation system.

02 Engineering the evidence chain

A result is useful only when its controller, protocol and data agree.



A completed run is not automatically a passed capability.

Three different completion states

- **Implemented:** source code and software checks exist.
- **Formally evaluated:** a frozen protocol produced a valid pass or negative result.
- **Development only:** a diagnostic may explain failure, but cannot authorize promotion.

Practical safeguards

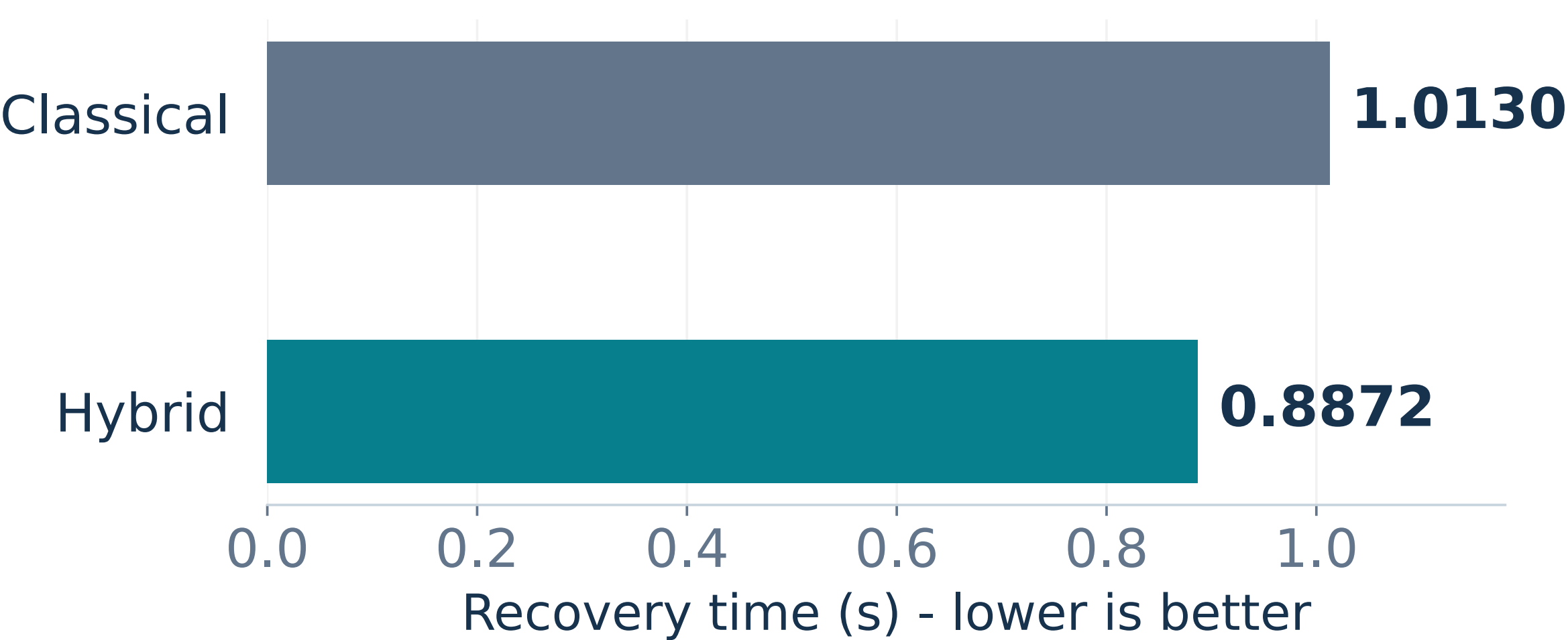
- **Do not substitute baselines.** Match commands, limits, artifacts and the historical source revision.
- **Measure the applied action.** Preserve channel order, signs, masks and saturation semantics.
- **Keep failed trials failed.** In RollBoundary, a single 5 ms bilateral unsupported substep invalidates the trial.
- **Preserve the evidence.** Archive gate records, checkpoint hashes and the development snapshot separately.

Role: research questions, integration, experiment design, diagnosis and evidence review. Implementation was collaborative and AI-assisted; sole authorship of every code path is not claimed.

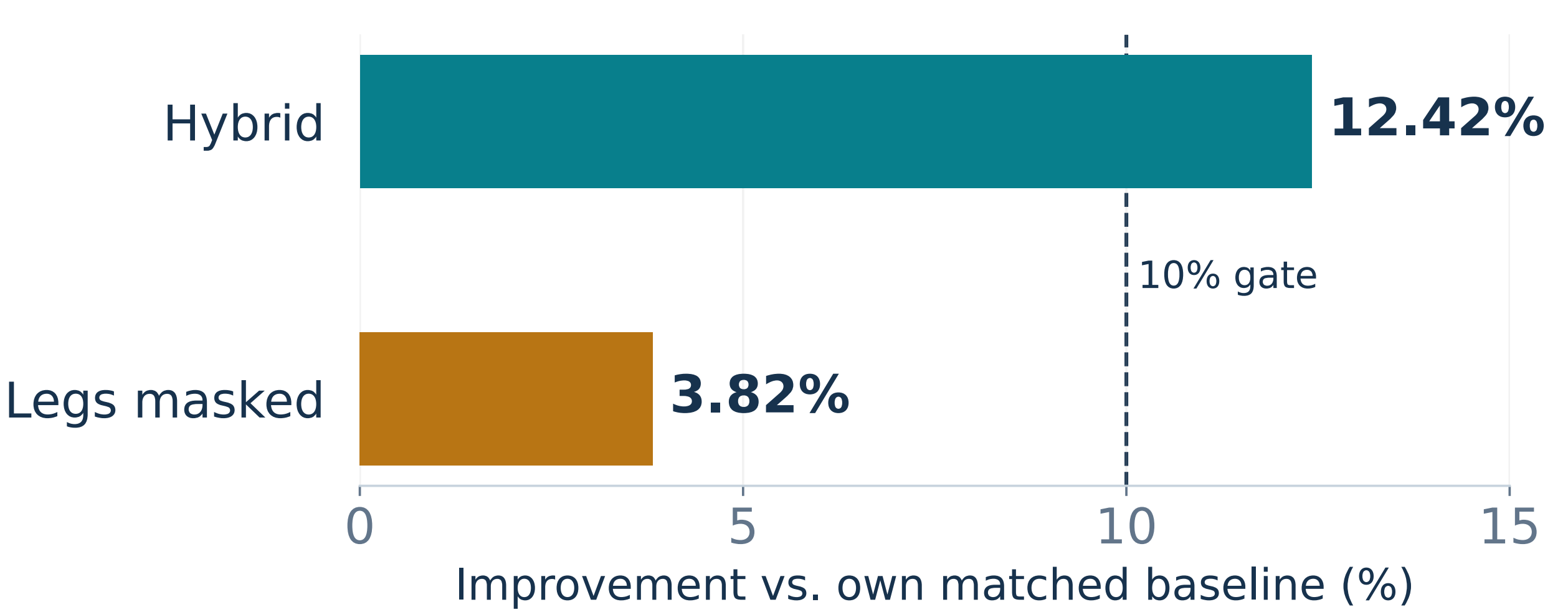
03 Representative validation

Residual value was measured, not assumed.

A Matched recovery comparison



B Evaluation-time leg ablation



Stage5 formal evidence, seed 1: 128 disturbance events per arm, zero event terminations. The ablation uses the same checkpoint with leg channels masked, and its own matched baseline. No cross-seed error bars or exact percentage attribution to the legs is claimed. [E6/A1]

A measured capability boundary

Later C1 qualification: 15/15 flat-control cells passed. [C1]

RollBoundary	Success	Strict outcome
Flat	96/96	qualified
2.5 mm step	18/96	not qualified

At 2.5 mm, 73/96 trials contained bilateral unsupported samples. Occasional traversal is not a safe capability claim. [R0]

Not established: reliable stair climbing, multi-seed generalization, or real-hardware closed-loop performance. Later contact/effort studies remain diagnostics, not formal upgrades.

Simulation only. Evidence IDs map to the report and SHA256 manifest. Foundations: Johannink et al. (ICRA 2019), Schulman et al. (2017), and Todorov et al. (IROS 2012).